VTPEH 6270 - Check Point 04

Blood Pressure Simulations

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1 Objective and Approach

1.1 Specific Objective

We will look at the impact of hypertension in men and women as they age. The CDC data I found gives only the rate of hypertensive disease (as opposed to actual counts) in adults 20 and older, per gender, poverty level, age group, race. We will focus on exploring hypertension disease in relation to gender and age.
* Does hypertension affect men and women in the same fashion?
* Has there been a change in the impact of hypertensive disease between the years 1990s and the late 2010s?

https://data.cdc.gov/National-Center-for-Health-Statistics/DQS-Hypertension-in-adults-age-20-and-older-by-sel/va5e-efw9/about_data

```
#set working directory and read data file
setwd("C:/Users/alexm/OneDrive/Documents/CLASSES/VTPEH6270/Check Point 04")
getwd()
data <-
  read.csv("data/Hypertension_in_adults_age_20_and_older.csv", header = TRUE)
```

```r
library(ggplot2)
library(dplyr)
library(tidyr)
library(tibble)
```

1.2 Data Description and Parsing

1.2.1 Parsing

The focus is on data that illustrates the impact of hypertensive disease by age group, gender and for two time periods, 1988-1994 & 2017-3/2020

```r
# simplify the data frame to keep only the data of interest
data_age <- subset(data, GROUP == "Sex and age group")

data_age_1990 <- subset(data_age, TIME_PERIOD == "1988-1994")
data_age_2017 <- subset(data_age, TIME_PERIOD == "2017-3/2020")

# remove columns of no use for this study
data_age_1990 <- data_age_1990[,c(1,8,11,13,14,15,16,18,20,21)]
data_age_2017 <- data_age_2017[,c(1,8,11,13,14,15,16,18,20,21)]

# remove redundant data
data_age_1990 <- subset(data_age_1990, !(SUBGROUP_ORDER %in% c(13,16,21,24)) )
data_age_2017 <- subset(data_age_2017, !(SUBGROUP_ORDER %in% c(13,16,21,24)) )

# create a numerical variable "Med_Age" to describe average age within each age group

data_age_1990 <- data_age_1990 %>%
   mutate(Med_Age = case_when(
     SUBGROUP_ORDER == 14 ~ 27,
     SUBGROUP_ORDER == 22 ~ 27,
     SUBGROUP_ORDER == 15 ~ 40,
     SUBGROUP_ORDER == 23 ~ 40,
     SUBGROUP_ORDER == 17 ~ 50,
     SUBGROUP_ORDER == 25 ~ 50,
     SUBGROUP_ORDER == 18 ~ 60,
     SUBGROUP_ORDER == 26 ~ 60,
     SUBGROUP_ORDER == 19 ~ 70,
     SUBGROUP_ORDER == 27 ~ 70,
     SUBGROUP_ORDER == 20 ~ 80,
     SUBGROUP_ORDER == 28 ~ 80
   ))

data_age_2017 <- data_age_2017 %>%
   mutate(Med_Age = case_when(
     SUBGROUP_ORDER == 14 ~ 27,
     SUBGROUP_ORDER == 22 ~ 27,
     SUBGROUP_ORDER == 15 ~ 40,
     SUBGROUP_ORDER == 23 ~ 40,
     SUBGROUP_ORDER == 17 ~ 50,
```

```
    SUBGROUP_ORDER == 25 ~ 50,
    SUBGROUP_ORDER == 18 ~ 60,
    SUBGROUP_ORDER == 26 ~ 60,
    SUBGROUP_ORDER == 19 ~ 70,
    SUBGROUP_ORDER == 27 ~ 70,
    SUBGROUP_ORDER == 20 ~ 80,
    SUBGROUP_ORDER == 28 ~ 80
  ))
```

1.2.2 Data Table

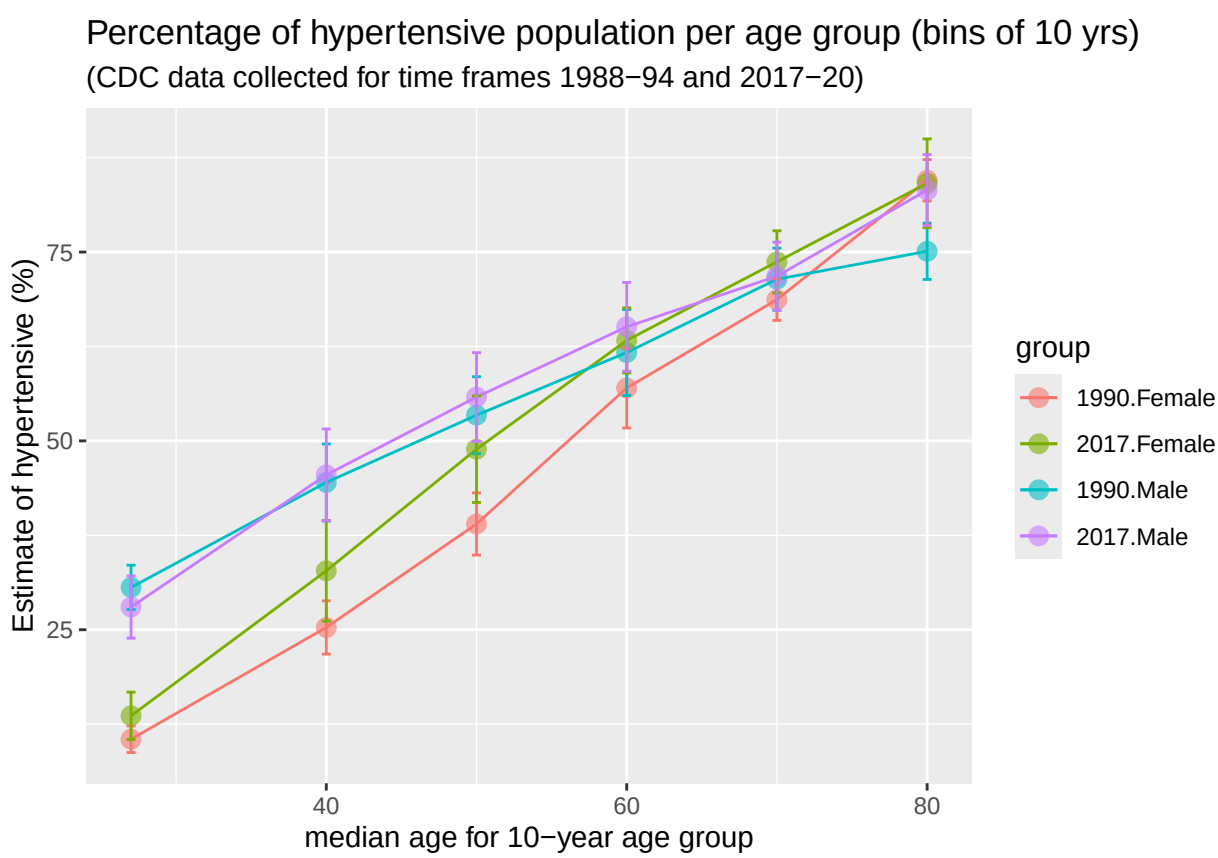
```
# table for the data

vars <- c("NESTING_LABEL", "Med_Age", "ESTIMATE", "STANDARD_ERROR")

# Function to convert R classes to human-readable types
get_var_type <- function(x) {
  if (is.numeric(x)) {
    "Continuous"
  } else if (is.factor(x) || is.character(x)) {
    "Categorical"
  } else {
    "Other"
  }
}

# Create summary table
variable_summary <- tibble(
  Variable = vars,
  Type = sapply(data_age_1990[vars], get_var_type),
  Class = sapply(data_age_1990[vars], class)
)

# View table
variable_summary
```

```
## # A tibble: 4 x 3
##   Variable       Type        Class
##   <chr>          <chr>       <chr>
## 1 NESTING_LABEL  Categorical character
## 2 Med_Age        Continuous  numeric
## 3 ESTIMATE       Continuous  numeric
## 4 STANDARD_ERROR Continuous  numeric
```

1.3 Data Visualization

```
# plot

data_age_1990$Year <- "1990"
```

```r
data_age_2017$Year <- "2017"

combined <- rbind(data_age_1990, data_age_2017)
combined <- combined %>%
  mutate(group = interaction(Year, NESTING_LABEL))

ggplot(combined,
       aes(x = Med_Age,
           y = ESTIMATE,
           color = group,
           group = group)) +
  geom_point(size = 3, alpha = 0.6) +
  geom_line() +
  geom_errorbar(aes(
    ymin = ESTIMATE - 1.96 * STANDARD_ERROR,
    ymax = ESTIMATE + 1.96 * STANDARD_ERROR
  ),
  width = 0.6) +
  labs( title= "Percentage of hypertensive population per age group (bins of 10 yrs)",
  subtitle="(CDC data collected for time frames 1988-94 and 2017-20)",
  x = "median age for 10-year age group",
  y = "Estimate of hypertensive (%)")
```



1.4 Plausible relationships:

From the plot, we make the following 2 observations:
* it seems that more young (20-35) women and/or men suffer from hypertension now compared to 25 years ago;
* women seem to be impacted by hypertension with a delay age-wise than men but then catch up with men around the age of 70.

```
aggregate(ESTIMATE ~ NESTING_LABEL, data = data_age_1990, FUN = mean)
aggregate(ESTIMATE ~ NESTING_LABEL, data = data_age_2017, FUN = mean)
```

- Are there more hypertensive young women and men in the recent time frame (2017-2020) than 25 years ago (1988-1994)?

- Is this trend statistically significant?
  We calculate z-scores for the 20-35 age group.

```
# calculating Z-scores for differences between data from 2017 and 1990, for age groups 20-35 for Women

p_Fem_1990_27 <- subset(data_age_1990, NESTING_LABEL == "Female" & data_age_1990$Med_Age == 27)$ESTIMATE
p_Fem_2017_27 <- subset(data_age_2017, NESTING_LABEL == "Female" & data_age_2017$Med_Age == 27)$ESTIMATE
p_Mal_1990_27 <- subset(data_age_1990, NESTING_LABEL == "Male" & data_age_1990$Med_Age == 27)$ESTIMATE
p_Mal_2017_27 <- subset(data_age_2017, NESTING_LABEL == "Male" & data_age_2017$Med_Age == 27)$ESTIMATE


se_Fem_1990_27 <- subset(data_age_1990, NESTING_LABEL == "Female" & data_age_1990$Med_Age == 27)$STANDAR
se_Fem_2017_27 <- subset(data_age_2017, NESTING_LABEL == "Female" & data_age_1990$Med_Age == 27)$STANDAR
se_Mal_1990_27 <- subset(data_age_1990, NESTING_LABEL == "Male" & data_age_1990$Med_Age == 27)$STANDARD_
se_Mal_2017_27 <- subset(data_age_2017, NESTING_LABEL == "Male" & data_age_1990$Med_Age == 27)$STANDARD_


z_27_Fem <- (p_Fem_2017_27 - p_Fem_1990_27) /
     sqrt(se_Fem_2017_27^2 + se_Fem_1990_27^2)

print(z_27_Fem)

p_value_27_Fem <- 2 * (1 - pnorm(abs(z_27_Fem)))

print(p_value_27_Fem)

z_27_Mal <- (p_Mal_2017_27 - p_Mal_1990_27) /
     sqrt(se_Mal_2017_27^2 + se_Mal_1990_27^2)

print(z_27_Mal)

p_value_27_Mal <- 2 * (1 - pnorm(abs(z_27_Mal)))

print(p_value_27_Mal)
```

The p-value is 0.09 for women and 0.31 for men for the age group 20-35. Hence the difference in mean observed between the 2 data sets is not statistically significant and H0 cannot be rejected.

Conclusion: There is no significant difference that we can detect in the percentage of hypertensive people between the years 1988-1994 and 2017-2020, for males or females in the age group 20-35.

- Are men suffering more from hypertension than women and
- does this female advantage decrease with age?

We will perform a linear regression:
Percent Hypertension ~ Gender + Age_Group + Gender x Age_Group

```
# linear regression with Interaction

# subset data for Females only and calculate parameters
data_age_1990_Fem <- subset(data_age_1990, NESTING_LABEL == "Female")
modelFem1990 <- lm(ESTIMATE ~ Med_Age, data = data_age_1990_Fem)
summary(modelFem1990)

data_age_1990_Mal <- subset(data_age_1990, NESTING_LABEL == "Male")
modelMal1990 <- lm(ESTIMATE ~ Med_Age, data = data_age_1990_Mal)
summary(modelMal1990)

model <- lm(ESTIMATE ~ NESTING_LABEL:Med_Age, data = data_age_1990)
summary(model)

model2 <- lm(ESTIMATE ~ NESTING_LABEL * Med_Age,
             data = data_age_2017)
summary(model2)
```

The discussed results or linear regressions are summarized in table 1.

Table 1: Table with linear regression parameters. Unit is mm Hg or mm Hg/year.

Variables	Male	Female
data origin	1984-1990	1984-1990
slope (a)	0.86	1.41
intercept (b)	9.43	-29.7
noise (RSE)	1.96	2.16

The p-values are less than 0.05.

Conclusion: The comparison between the intercept data for women and men indicates that women start at a young age with a lower rate of hypertension than men (although the absolute numbers make no sense as newborns do not have negative blood pressure), but the slope of increase in the rate of hypertension over age is steeper for women and thus their rate of hypertension catches up with men at a later age. Those differences are significant ($p<0.05$). The noise (RSE) is low compared to the signal (increase of 0.86 (males) or 1.41 (females) units (Mm Hg) per year.)

2 Simulation to explore group size

2.1 Simulation of statistics for groups of uneven sizes

For the first simulation, we will simulate data based on the statistics published in Suaarez-Gonzalez et al, 2025 "Relationships between diet, sociodemographic factors, and body composition in students from UNEMI". In particular, we will simulate data to match the means published in Table 2:
Women- nutrient rich diet (NRD- categorical variable- either Good or Bad) and systolic blood pressure (SBP, numerical) with means(sd) reported in the article as 105.6(7.1) for Good NRD and 113.9(9.3) for Bad NRD. Units for all SBP are mm Hg. The article reported a very low percentage of subjects with Good NRD (2%) and I am curious to see how the authors were able to see a significant association between Good NRD and SBP with a p-value reported of 0.000.

We create a data frame with 250 observations and a probability of 2% of subjects with Good NRD.

```
# Set the seed for reproducibility
set.seed(123)

# Number of observations
nn <- 250

# Probability of Nutrient Rich Diet consumption
prob_NRD_Good <- 0.02

# Simulate data
data <- data.frame(
  NRD = sample(c("Good", "Bad"), nn, replace = TRUE, prob = c(prob_NRD_Good, 1 - prob_NRD_Good))
)

# Display the first few rows of the simulated dataset
head(data)

# count the number of rows with Good NRD
sum(data$NRD == "Good")
```

We set the mean and sd at the valued published in the article and assign the SBP values to match the means and sd.

```
# Mean Systolic Blood Pressure score in each group
SBP_NRD_Good = 105.6
SBP_NRD_Bad = 113.9
SBP_sd = 8
data$SBP = NA
Effect = 8.3

for (ii in 1:nrow(data)){
  if (data$NRD[ii] == "Good"){
    data$SBP[ii] = rnorm(1,
                                SBP_NRD_Good,
                                SBP_sd)
```

```
  }else if (data$NRD[ii] == "Bad"){
    data$SBP[ii] = rnorm(1,
                                SBP_NRD_Good+Effect,
                                SBP_sd)
  }
}
```

2.1.1 Simulation output

We verify using the aggregate function that the means and sd match the constraints we placed on the simulation.

```
# Summary table
aggregate(SBP ~ NRD, data, FUN = mean)
aggregate(SBP ~ NRD, data, FUN = sd)
```

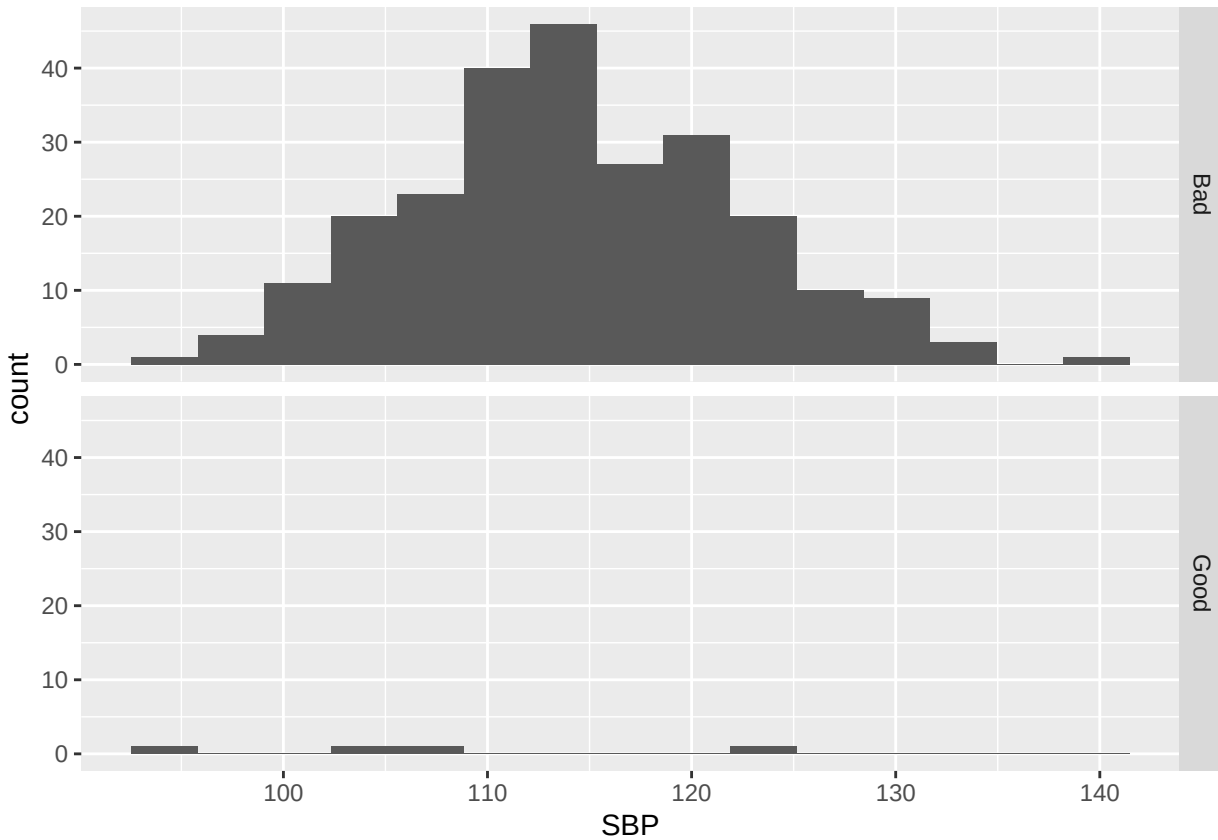
2.1.2 Visual check:

The low number of subjects in the "Good" NRD group is only 4, hence the histogram, that looks abnormal.

2.1.3 Interpretation

In conclusion, a study with such a small number of subjects in a group is cause for concern.

```
ggplot(data, aes(x = SBP)) +
  geom_histogram(bins = 15) +
  facet_grid(NRD ~.)
```



3 Simulation to explore power

We now explore impact of effect, noise and sample size on statistical power.

3.1 Identify parameters

We will use the following parameters:

- n (nn): sample size
- p_c (prob_NRD_Good): probability of Nutrient Rich diet (NRD_Good)
- h_c (SBP_NRD_Good): Systolic Blood Pressure given Nutrient Rich diet
- $h_{\overline{c}}$ (SBP_NRD_Bad): Systolic Blood Pressure with Nutrient Poor diet- is SBP_NRD_Good + Effect
- h_{sd} (SBP_sd): standard deviation of Systolic Blood Pressure within each group

3.2 Function

```
f_sim_SBP = function(nn,
                        prob_NRD_Good,
                        SBP_NRD_Good,
                        SBP_NRD_Bad,
```

```
                        SBP_sd
                        ){

  # Nutrient Rich Diet vector
  v_NRD = sample(
    c("Good", "Bad"),
    nn,
    replace = TRUE,
    prob = c(prob_NRD_Good, 1 - prob_NRD_Good)
  )

  # Systolic Blood Pressure vector -
  v_SBP = NA
  for (jj in 1:nn) {
    if (v_NRD[jj] == "Good") {
      v_SBP[jj] = rnorm(1, SBP_NRD_Good, SBP_sd)
    } else if (v_NRD[jj] == "Bad") {
      v_SBP[jj] = rnorm(1, SBP_NRD_Bad, SBP_sd)
    }
  }

  # Combine data
  data = data.frame(
    NRD = v_NRD,
    SBP = v_SBP)

  return(data)
}
```

3.3 Simulation runs

We use 4 `for` loops and store the outputs in `list()`
The 4 loops are:
* 3 values of noise
* 10 valued of effect
* 10 samples sizes
and for each combination, we run 100 simulations.
Note that the 3 values of noise tested are 40%, 80% and 120% of the maximum effect value explored.

```
# Parameters for tests
# prob_NRD_good = 0.4
# SBP_RND_Good = 105.6
# SBP_NRD_Bad = 113.9
# SBP_sd = 8

# Initialize all objects: lists, data frame holding statistics
list_datasets <- list()
ii = 0
nn = 0

df_stats <- data.frame(beta = NA,
```

```r
                        SBP_sd = NA,
                        sample = NA,
                        sim = NA,
                        SBP_GOOD_mean = NA,
                        SBP_GOOD_sd = NA,
                        SBP_BAD_mean = NA,
                        SBP_BAD_sd = NA,
                        p = NA)
sim_index = 1
SBP_NRD_Good = 105.6

# Loop on noise
for (SBP_sd in seq(4, 12, 4)) {

# Loop on Effect
for (beta in seq(0, 10, 1)) {

  SBP_NRD_Bad = SBP_NRD_Good + beta

  # Loop on sample size
  for (nn in seq(25, 275, 25)) {

    # Run simulation 100 times
    for (ii in 1:100) {

      # Generate data
      sim_data <- f_sim_SBP(
        nn = nn,
        prob_NRD_Good = 0.4,
        SBP_NRD_Good = SBP_NRD_Good,
        SBP_NRD_Bad = SBP_NRD_Bad,
        SBP_sd = SBP_sd
      )

      # Store full dataset (comment to go faster)
      #list_datasets[[sim_index]] <- sim_data

      # Calculate mean and sd for each cheese group
      df_stats[sim_index, ] = NA
      df_stats[sim_index, "beta"] = beta
      df_stats[sim_index, "sample"] = nn
      df_stats[sim_index, "noise"] = SBP_sd
      df_stats[sim_index, "sim"] = ii
      df_stats$SBP_NRD_Good_mean[sim_index] =
        round(mean(sim_data[sim_data$NRD == "Good", "SBP"]), 2)
      df_stats$SBP_NRD_Good_sd[sim_index] =
        round(sd(sim_data[sim_data$NRD == "Good", "SBP"]), 2)
      df_stats$SBP_NRD_Bad_mean[sim_index] =
        round(mean(sim_data[sim_data$NRD == "Bad", "SBP"]), 2)
      df_stats$SBP_NRD_Bad_sd[sim_index] =
        round(sd(sim_data[sim_data$NRD == "Bad", "SBP"]), 2)

      # T-test
```

```
      test = t.test(SBP ~ NRD, sim_data)
      df_stats$p[sim_index] <- test$p.value

      # ???
      sim_index = sim_index + 1

    }

  }

}

}

# Clean environment
remove(ii, nn, beta, SBP_sd, sim_index, test, sim_data)

# Check: 1 simulation run
#head(list_datasets[[1]])

# Check: compiled summary statistics
head(df_stats)

# Save simulation summary
write.csv(df_stats, "sim_summary.csv", row.names = F, quote = F)
```

Create a pivot table with results

```
# Pivot stats
df_stats_long <- df_stats %>%
  select(beta, sample, noise, sim, SBP_NRD_Good_mean, SBP_NRD_Bad_mean) %>%
  pivot_longer(cols = c(SBP_NRD_Good_mean, SBP_NRD_Bad_mean),
               names_to = "group",
               values_to = "mean_value") %>%
  mutate(group = ifelse(group == "SBP_NRD_mean", "Good", "Bad"))
```

3.4 Visualisation of the power of the test for different 3 different noise levels

Scatter plots and Heat maps for 3 different levels of noise.

```
# scatter plot with noise as 3rd dimension. Sample size values are not visualized.
p4= ggplot(df_stats, aes(x = beta, y = p, color = noise)) +
  geom_jitter() +
  labs(x = "Effect size",
       y = "p-value") +
  scale_color_gradient(name = "Noise") +
  theme_minimal()
p4
```

```r
# scatter plot with sample size as 3rd dimension. Noise values are not visualized.
p5= ggplot(df_stats, aes(x = beta, y = p, color = sample)) +
  geom_jitter() +
  labs(x = "Effect size",
       y = "p-value") +
  scale_color_gradient(name = "Sample size") +
  theme_minimal()

p5
```

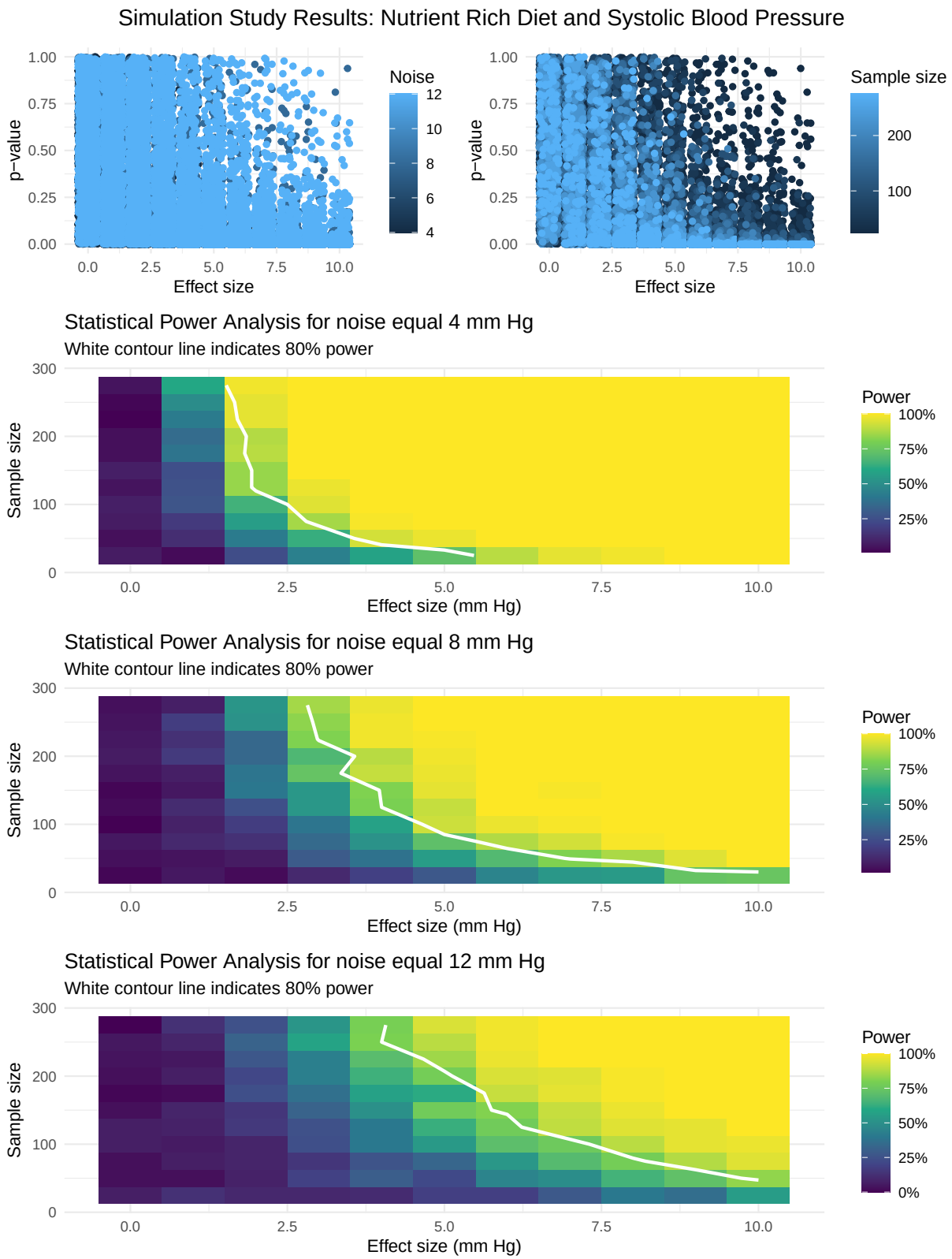
```r
# subset the result table in 3 tables for the 3 different noise levels
df_stats4 <- subset(df_stats, noise == "4")
df_stats8 <- subset(df_stats, noise == "8")
df_stats12 <- subset(df_stats, noise == "12")

# table and plot for noise level = 4
power_results4 <- df_stats4 %>%
  group_by(beta, sample) %>%
  summarise(
    power = sum(p < 0.05, na.rm = TRUE)/sum(!is.na(p)),
    .groups = "drop"
  )

p1= ggplot(power_results4, aes(x = beta, y = sample, fill = power)) +
  geom_tile() +
  scale_fill_viridis_c(name = "Power",
                       labels = scales::percent_format()) +
  geom_contour(aes(z = power), breaks = 0.8, color = "white", size = 1) +
  labs(x = "Effect size (mm Hg)",
       y = "Sample size",
       title = "Statistical Power Analysis for noise equal 4 mm Hg",
       subtitle = "White contour line indicates 80% power") +  # Fixed subtitle
  theme_minimal()
p1
```

```r
# table and plot for noise level = 8
power_results8 <- df_stats8 %>%
  group_by(beta, sample) %>%
  summarise(
    power = sum(p < 0.05, na.rm = TRUE)/sum(!is.na(p)),
    .groups = "drop"
  )

p2= ggplot(power_results8, aes(x = beta, y = sample, fill = power)) +
  geom_tile() +
  scale_fill_viridis_c(name = "Power",
                       labels = scales::percent_format()) +
  geom_contour(aes(z = power), breaks = 0.8, color = "white", size = 1) +
  labs(x = "Effect size (mm Hg)",
       y = "Sample size",
       title = "Statistical Power Analysis for noise equal 8 mm Hg",
       subtitle = "White contour line indicates 80% power") +  # Fixed subtitle
  theme_minimal()
```

```r
p2
```

```r
# table and plot for noise level = 12

power_results12 <- df_stats12 %>%
  group_by(beta, sample) %>%
  summarise(
    power = sum(p < 0.05, na.rm = TRUE)/sum(!is.na(p)),
    .groups = "drop"
  )

p3= ggplot(power_results12, aes(x = beta, y = sample, fill = power)) +
  geom_tile() +
  scale_fill_viridis_c(name = "Power",
                       labels = scales::percent_format()) +
  geom_contour(aes(z = power), breaks = 0.8, color = "white", size = 1) +
  labs(x = "Effect size (mm Hg)",
       y = "Sample size",
       title = "Statistical Power Analysis for noise equal 12 mm Hg",
       subtitle = "White contour line indicates 80% power") +  # Fixed subtitle
  theme_minimal()
p3
```

```r
# Compose plot
library (ggplot2)
library(patchwork)

combined_plot <- (p4 |p5) / (p1) /  (p2)  /  (p3) +
  plot_annotation(
    title = "Simulation Study Results: Nutrient Rich Diet and Systolic Blood Pressure",
    theme = theme(plot.title = element_text(size = 14, hjust = 0.5))
  )

combined_plot
```


Simulation Study Results: Nutrient Rich Diet and Systolic Blood Pressure
p−value
Effect size
Noise
Sample size
Statistical Power Analysis for noise equal 4 mm Hg
White contour line indicates 80% power
Statistical Power Analysis for noise equal 8 mm Hg
White contour line indicates 80% power
Statistical Power Analysis for noise equal 12 mm Hg
White contour line indicates 80% power
Sample size
Effect size (mm Hg)
Power

3.5 Interpretation

The heat maps for the different noise levels give results that surprise me: for a noise level superior to the effect size, the signal can still be detected as long as the sample size is large enough. For example, for an effect size of 7.5 mm Hg and noise level of 12 mm Hg, a significant association can be detected with a sample size above ~80. This seems counter intuitive to me.